



DESIGN AND CONTROL OF A MULTI-DOF ROBOTIC ARM USING WIRED CONTROLLER

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Abstract—The development of robotic arms has significantly contributed to automation across various industrial and research applications. This paper presents the design and implementation of a 5 Degrees of Freedom (DOF) Robotic Arm capable of performing complex tasks with precision and flexibility. The system utilizes servo motors for joint actuation and a microcontroller-based control unit for coordinated motion. The arm is designed to mimic human arm movements, allowing it to perform pick-and-place operations, object sorting, and precise positioning. Kinematic analysis, both forward and inverse, has been conducted to achieve accurate control over the end-effector. The proposed design aims to provide a low-cost, efficient, and easily programmable robotic solution suitable for laboratory automation, educational purposes, and light industrial tasks. Experimental results demonstrate smooth and accurate motion of the robotic arm, validating the design's reliability and effectiveness. The design incorporates both forward and inverse kinematic analysis to accurately determine joint angles and end-effector positions, ensuring smooth and precise operations. The arm is capable of performing pick-and-place operations, object sorting, and basic assembly tasks, making it suitable for laboratory automation, educational demonstrations, and light industrial applications. The paper also discusses the integration of sensors for feedback control and obstacle avoidance, improving the arm's adaptability and operational safety. Experimental results demonstrate the robotic arm's accuracy, repeatability, and efficiency, highlighting its potential as a low-cost, versatile solution for automation challenges.

1. INTRODUCTION

Robotics has emerged as a transformative technology in industries, research laboratories, and educational institutions due to its ability to perform tasks with high precision, consistency, and efficiency. Among robotic systems, robotic arms are widely used for applications such as **material handling, assembly, packaging, and laboratory automation**. The development of a robotic arm with multiple degrees of freedom allows it to mimic human arm movements, enabling complex tasks that require dexterity and flexibility. These devices utilize electrical impulses to incapacitate a target temporarily, typically by disrupting the motor control of muscles and causing involuntary spasms. Their popularity is due in part to their portability, ease of use. This research focuses on the design and implementation of a **5 Degrees of Freedom (DOF) Robotic Arm**, which provides a balance between complexity, functionality, and cost-effectiveness. The 5 DOF configuration allows the arm to perform movements such as rotation, lifting, and object

manipulation while maintaining precise control over the end-effector's position and orientation. The system employs **servo motors** for joint actuation and a **microcontroller-based control unit** for real-time coordination, motion planning, and execution of tasks.

Kinematic analysis, including **forward and inverse kinematics**, is applied to accurately determine the joint angles for desired end-effector positions. The integration of sensors further enhances the arm's performance by enabling feedback control and improving operational safety. This project aims to provide a low-cost, flexible, and easily programmable robotic solution that can serve educational purposes, light industrial applications, and experimental automation tasks.

The design, implementation, and experimental results discussed in this paper demonstrate the robotic arm's capability to perform precise and repeatable operations, highlighting its potential as a practical solution for modern automation challenges.

The primary objective of this research is to **design, develop, and implement a 5 DOF robotic arm** that can execute tasks with high accuracy and efficiency. The arm is powered by **high-torque servo motors** and controlled by a **microcontroller-based system**, which enables real-time motion control, path planning, and task execution. The system also integrates sensors for feedback control, obstacle detection, and operational safety, enhancing its reliability and adaptability in dynamic environments.

A critical aspect of robotic arm development is **kinematic analysis**, including forward and inverse kinematics, which allows precise calculation of joint angles and end-effector positions for performing desired tasks. The robotic arm presented in this study is capable of performing pick-and-place operations, object sorting, basic assembly, and other automation-related tasks.

This research aims to provide a **low-cost, flexible, and easily programmable robotic solution** that can be used for practical automation challenges, educational demonstrations, and laboratory experiments. Experimental results demonstrate the arm's smooth operation, repeatability, and accuracy, confirming its effectiveness as a functional robotic system. By integrating mechanical design, electronics, and control algorithms, this project contributes to the development of efficient, adaptable, and intelligent robotic systems.

2. Principles of Operation

- a. The operation of a **5 Degrees of Freedom (DOF) robotic arm** is based on the **integration of mechanical, electrical, and control systems**, which work together to achieve precise, human-like movements. The main principle revolves around **joint actuation, kinematic control, and coordinated motion of multiple degrees of freedom** to manipulate objects effectively. Current Flow and Neuromuscular Disruption

1. Mechanical Principle

The robotic arm consists of **five revolute joints**, each providing a rotational degree of freedom. These joints are powered by **servo motors**, which convert electrical signals into controlled mechanical motion. The segments of the arm are connected by rigid links, forming a kinematic chain. The movement of the arm depends on the coordinated rotation of these joints to position the **end-effector** (gripper or tool) at a desired location with the required orientation.



2. Electrical and Control Principle

The **microcontroller** serves as the central processing unit, controlling all servo motors based on the input commands. The system receives commands from either a **pre-programmed routine, a joystick, or a computer interface**, which are converted into **pulse-width modulation (PWM) signals** for the servo motors. Each PWM signal determines the angular position of the corresponding servo motor, enabling precise control of each joint.

3. Kinematic Analysis Principle

To achieve accurate motion, the arm relies on **kinematic equations**:

- **Forward Kinematics:** Determines the position and orientation of the end-effector based on the known angles of each joint. This helps predict where the arm will reach when the joint angles are set.
- **Inverse Kinematics:** Calculates the necessary joint angles to place the end-effector at a desired position and orientation. This is essential for task execution such as pick-and-place operations.

4. Operational Workflow

1. **Task Input:** The desired position and orientation of the end-effector are defined.
2. **Inverse Kinematics Calculation:**

The microcontroller computes the required joint angles.

3. **Servo Motor Actuation:** PWM signals drive the servo motors to the calculated angles.

4. **Motion Execution:** The arm moves smoothly along the trajectory to reach the target.

- **Feedback and Adjustment:** Optional sensors **Safety Mechanisms:** Modern stun devices incorporate safety features such as safety switches or buttons that prevent accidental discharge. These mechanisms ensure that the device is only activated when the user intends to use it.

- **Compactness:** The portability of electric stun devices is crucial, especially for personal defense. These devices are designed to be lightweight and easy to carry, often fitting in a purse or pocket.

5. (such as limit switches or encoders) provide feedback to correct any positional errors and ensure precise operation.

between the brain and muscles, leading to a temporary loss of voluntary control over movement.

5. Automation and Control Principle

The robotic arm's operation is a combination of **open-loop and closed-loop control**:

- **Open-Loop Control:** Moves the arm based on pre-defined commands without feedback.
- **Closed-Loop Control (Optional):** Uses sensors to monitor position and correct errors, increasing accuracy and reliability.

In summary, the **5 DOF robotic arm operates by transforming electrical control signals into precise mechanical motion**, coordinating multiple joints to manipulate objects with human-like dexterity. The integration of kinematic analysis, servo actuation, and feedback control ensures smooth, accurate, and repeatable operations, making it suitable for automation, educational, and research applications.

4. Operational Workflow

1. **Task Input:** Desired end-effector position and orientation are defined.
2. **Inverse Kinematics:** The microcontroller calculates the joint angles needed.
3. **Servo Actuation:** PWM signals move the joints to the calculated angles.
4. **Motion Execution:** The arm follows a smooth trajectory to reach the target.
5. **Feedback Control (Optional):** Sensors provide positional feedback, allowing error correction for precise operation.

5. Automation and Control Principle open-loop and closed-loop control

- **Open-Loop:** Executes pre-defined movements without feedback.
- **Closed-Loop:** Uses sensors to monitor and correct joint positions, improving accuracy and reliability.

6. Safety and Efficiency Principles

- Integration of **limit switches and sensors** prevents the arm from exceeding joint limits or colliding with objects.
- Efficient servo motor selection and optimized kinematic planning reduce energy consumption and improve response time.

Electrical and Control Principle

A **microcontroller** acts as the central processing unit,

receiving commands from a joystick, computer, or pre-programmed routine. These commands are converted into **PWM (Pulse Width Modulation) signals**, which control the angle of each servo motor. This ensures accurate and synchronized movement of all joints.

3. Applications of Electric Stun Gun

1. Industrial Applications

- **Material Handling:** Pick-and-place operations on assembly lines, moving products from one location to another.
- **Packaging:** Automated packaging of products in factories, improving speed and efficiency.
- **Assembly Tasks:** Performing repetitive assembly operations, such as screwing, fitting, or sorting components.
- **Painting and Welding:** Can be programmed to perform uniform painting or spot welding in production lines.

2. Educational and Research Applications

- **Learning Tool:** Helps engineering students understand robotics, kinematics, and automation principles.
- **Laboratory Experiments:** Used to study control algorithms, path planning, and robotic manipulation.
- **Simulation of Industrial Processes:** Students can test robotic workflows in a safe, controlled environment.

3. Medical and Rehabilitation Applications

- **Surgical Assistance:** Can assist in precision tasks during surgeries (though actual surgical robots usually have more DOF).
- **Rehabilitation Therapy:** Assists patients in physiotherapy by providing repetitive, controlled movements

4. Laboratory Automation

- **Sample Handling:** Moves test tubes, Petri dishes, or chemical samples in laboratories.
- **Experiment Automation:** Performs repetitive laboratory tasks to save time and reduce human error.

5. Pick-and-Place and Sorting Applications

- Sorting items by size, shape, or color.
- Handling fragile items where precise positioning is essential.

vary based on several factors, including the duration of exposure, the area of the body affected, and the overall health of the individual receiving the shock.

a. Physiological Effects

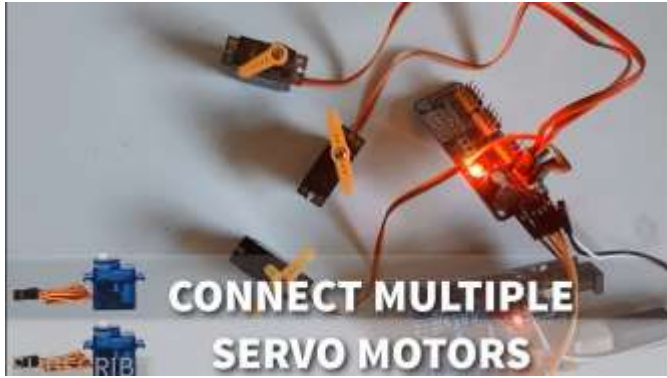
The primary physiological effect of an electric stun device is the disruption of the body's neuromuscular system. The current causes involuntary muscle spasms and

ROBO ARM MODEL



The 5 Degrees of Freedom (DOF) robotic arm is designed to replicate the movements of a human arm, providing flexibility, precision, and versatility in handling objects. The mechanical structure consists of rigid links connected by five revolute joints, forming a kinematic chain. The base serves as the foundation of the arm, supporting all other components and containing a servo motor for rotational motion. The shoulder link connects the base to the upper arm and allows vertical movement, while the elbow link enables bending and extension of the forearm. The wrist, comprising two joints, controls the pitch and roll of the end-effector, which is usually a gripper or tool capable of picking, placing, or manipulating objects. Lightweight and durable materials such as aluminum, acrylic, or high-strength plastic are used for the links, and bearings are incorporated in the joints to reduce friction and ensure smooth motion. Each joint is actuated by a servo motor, with high-torque servos used for the base and shoulder to handle the weight of the arm, and smaller servos for the wrist and gripper for precise movements. The arm is controlled by a microcontroller, which receives commands and sends PWM signals to the servo motors to achieve accurate joint rotation. A stable power supply is required to run the motors, and optional sensors such as encoders or limit switches provide feedback for improved accuracy and operational safety. The assembly process involves fixing the base on a stable platform, installing servo motors at each joint, connecting the links with screws and brackets, attaching the end-effector, wiring the electronics to the microcontroller and power supply, and calibrating each joint for coordinated movement. When operational, the microcontroller sends signals to the servo motors, and the coordinated movement of all five joints positions the end-effector in 3D space to perform tasks such as pick-and-place, sorting, or light assembly. The modular and cost-effective design of the 5 DOF robotic arm makes it ideal for educational purposes, laboratory experiments, research projects, and light industrial automation.

9. PIN CONFIGURATION



Summary (5 servo channels + comms + sensors):

- **Servo 1 (Base rotation)** — TIM2_CH1 → PA0
- **Servo 2 (Shoulder)** — TIM2_CH2 → PA1
- **Servo 3 (Elbow)** — TIM2_CH3 → PA2
- **Servo 4 (Wrist pitch)** — TIM2_CH4 → PA3
- **Servo 5 (Wrist roll / gripper orient)** — TIM3_CH1 → PA6
- **Servo 6 (PickUp)** — TIM2_CH5 → PA2

10. Hardware Required

- └ STM32F103C8T6 (“Blue Pill”) microcontroller
- └ High-torque servo motor for base joint
- └ High-torque servo motor for shoulder
- └ joint Standard servo motor for elbow
- └ joint Standard servo motor for wrist
- └ pitch joint Small servo motor for wrist
- └ roll / gripper 5V regulated power supply
- └ for servo motors Power supply for
- └ STM32 microcontroller Jumper wires / connecting cables
- └ Decoupling capacitors (0.1 μ F for MCU pins)
- └ Bulk capacitor (1000 μ F for servo power stabilization)
- └ Limit switches (2–4 pieces) for joint safety

FEATURES OF EACH COMPONENT :

STM32F103C8T6 Microcontroller

The STM32F103C8T6 is a 32-bit ARM Cortex-M3 microcontroller that acts as the brain of the robotic arm. Operating at 3.3V with a clock speed of 72 MHz, it provides multiple PWM channels to control all five servo motors accurately. The microcontroller also supports communication protocols like USART, I2C, and SPI, enabling integration with sensors, modules, and external devices.

Its low power consumption, compact size, and high processing speed make it ideal for precise motion control and real-time operations in robotic systems.

High-Torque Servo Motors (Base and Shoulder)

The base and shoulder joints use high-torque servo motors capable of producing 10–20 kg·cm of torque, which is sufficient to lift the arm and its payload. These servos convert PWM signals from the microcontroller into precise rotational motion. Their robust design ensures stability and smooth operation, while the high torque allows the arm to perform lifting and rotating movements without strain.

Standard Servo Motors (Elbow and Wrist Pitch)

The elbow and wrist pitch joints utilize standard servo motors that provide controlled rotation with moderate torque suitable for mid-weight links. These motors are compact and respond quickly to PWM commands, ensuring smooth and accurate movement of the arm. They are optimized for speed and precision, allowing the arm to perform coordinated motions efficiently.

Small Servo Motor (Wrist Roll / Gripper)

The wrist roll and gripper are powered by small servo motors that provide lower torque (3–5 kg·cm) appropriate for the end-effector. Despite their compact size, these servos deliver precise rotational control, allowing the gripper to pick, place, or manipulate objects accurately. Their lightweight design minimizes load on the wrist joint and helps maintain the overall balance of the arm.

Gripper / End-Effector

The gripper, or end-effector, is the component responsible for interacting with objects. It can be a two-fingered claw, a specialized tool, or a custom-designed mechanism. The gripper's force depends on the servo motor used, and it is designed to be lightweight to avoid overloading the wrist joint. This component is crucial for performing tasks like pick-and-place operations, object handling, or light assembly.

Rigid Links

The five rigid links connect the joints and form the arm's kinematic chain. They are typically made from aluminum, acrylic, or high-strength plastic to provide a balance between durability and lightweight construction. These links ensure the arm maintains its structural stability while allowing smooth and accurate motion across all joints.

Brackets, Screws, and Ball Bearings

Brackets and screws are used to assemble the links and secure servo motors in place, ensuring proper alignment and stability. Ball bearings at the joints reduce friction, allowing smooth rotation and prolonging the lifespan of both the servos and mechanical components. Together, these mechanical elements ensure reliable and precise movements.

Power Supplies

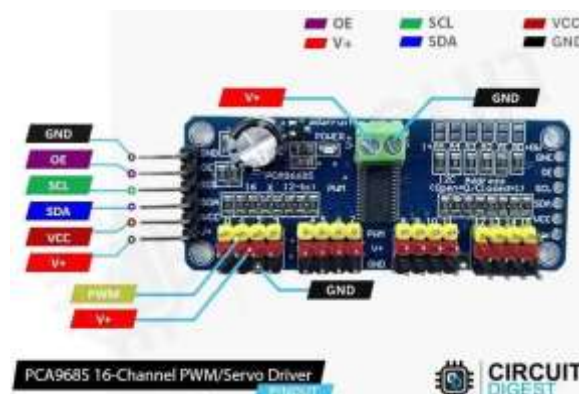
The robotic arm requires two power sources: a 5V regulated supply for the servo motors and a 3.3V or 5V supply for the STM32 microcontroller. The servo power supply must handle peak currents to prevent voltage drops, while the microcontroller supply ensures stable operation of the control system. Bulk capacitors are often added to smooth voltage fluctuations and prevent servo jitter.

Decoupling and Bulk Capacitors

Decoupling capacitors (0.1 μF) are used on the MCU pins to reduce voltage spikes, while bulk capacitors (1000 μF) are connected to the servo power supply to stabilize voltage during sudden current draws. These capacitors improve the reliability and smooth operation of the entire system.

CONCLUSION

11. The design, development, and implementation of a 5 Degrees of Freedom (DOF) robotic arm demonstrate the integration of mechanical design, electronics, and control systems to achieve precise and versatile automation. This project highlights how a robotic arm with five joints can mimic human-like movements, enabling tasks such as pick-and-place operations, object manipulation, sorting, and light assembly. The combination of high-torque servo motors, a microcontroller-based control system, and optional sensors provides accurate motion, repeatability, and operational safety. Through kinematic analysis, including forward and inverse kinematics, the arm is capable of determining the precise joint angles required to move the end-effector to a desired position, ensuring efficiency and reliability in task execution. The modular and cost-effective design of the arm makes it suitable not only for educational and laboratory purposes but also for small-scale industrial applications where repetitive tasks require precision and consistency. By integrating components such as servo motors, limit switches, and feedback sensors, the system achieves a balance between functionality, stability, and safety. Furthermore, the project demonstrates the practical application of concepts such as PWM control, sensor feedback, and microcontroller programming, offering valuable insights into modern robotics. It provides a foundation for further enhancements, such as adding additional degrees of freedom, implementing wireless control, integrating advanced sensors, or programming complex task sequences. The successful realization of this project illustrates the potential of robotics in transforming industrial, educational, and research environments by improving efficiency, accuracy, and productivity,



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